Chapter 0
Section 0.3
definations, Theorems and Proofs

Definations
Describe the objects and notions that we use.

Mathematical Statements Expresses that some objects has a certain property

Proof: Convincing logical argument that a statement is true

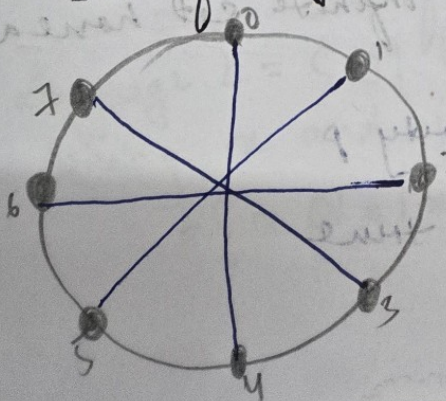
Theorm: A mathematical statement proved true

Types of Proofs
1) Proofs by construction
2) " " contradiction
3) " " induction

Proofs by Construction — Proof comes very often
Prove a Theorem by demonstrating how to construct a object.

Theorm
For each even numbers n greater than 2, there exists a 8-regular graph with n nodes.
k-regular graph ⇐ every node has a degree k.

Proof: Graph $G_1 = (V, E)$
Set of nodes $V = \{0, 1, 2 \dots i \dots n-1\}$
Set of edges $E = \{\{i, i+1\} \mid \text{for } 0 \le i < n-2\} \cup \{\{n-1, 0\}\} \cup \{\{i, i+\frac{n}{2}\} \mid \text{for } 0 \le i \le \frac{n}{2} - 1\}$



Let $n = 8$

i	$i + \frac{n}{2} = i + \frac{8}{2} = i+4$
0	$0+4 = 4$
1	$1+4 = 5$
2	$2+4 = 6$
3	$3+4 = 7$

Proof by Contradiction

Assume that a theorem is false
Then show that is assumption leads to an obviously false consequence called a contradiction.

Theorem

$\sqrt{2}$ is irrational.

Proof: Assume $\sqrt{2}$ is rational

$\sqrt{2} = \frac{m}{n}$. If both m and n are divisible by the same integer greater than 1, divide both by the largest such integer. At least one of m and n must be an odd number.

$n\sqrt{2} = m$

$2n^2 = m^2 \Rightarrow$ m is even $\quad m = 2K$

$2n^2 = (2K)^2 = 4K^2$

$n^2 = 2K^2 \longrightarrow n$ is even

Both m and n are even — a contradiction

Proof by Induction

Show that all elements of an infinite set have a specified property

$P(K) \leftarrow$ elements K has property P

Basis: Prove that $P(1)$ is true

Induction step: For each $i \geq 1$ - assume that $P(i)$ is true and use this assumption to show that $P(i+1)$ is true

Chapter - 1

Regular Languages

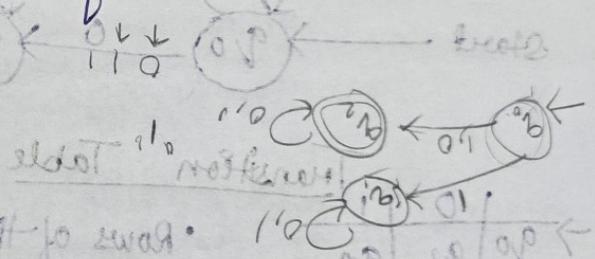
Regular Languages are those languages that can be described by finite automata. machine with limited memory

A finite automaton has a set of states and its control moves from state to state in response to external inputs.

Deterministic Finite Automata (DFA) or Finite State Machine (FSM)

DFA is in a single state after reading any sequence of inputs.

On each input there is only one state to which the automaton can transition from its current state



singular : Automaton

Plural : Automata

Defination of DFA

$$A = (Q, \Sigma, \delta, q_0, F)$$

$A \leftarrow$ name of DFA

$Q \leftarrow$ finite set of states

$\Sigma \leftarrow$ finite set of input symbols

$\delta \leftarrow$ transition function. $\delta : Q \times \Sigma \rightarrow Q$

$q_0 \leftarrow$ start state, $q_0 \in Q$

$F \leftarrow$ set of final or accepting states $F \subseteq Q$.

Language $L = \{w \mid w$ is of the form $x01y$ for some string x and y consisting of 0's and 1's only$\}$

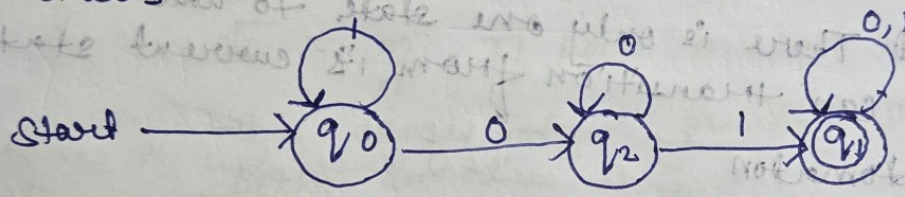
$= \{x01y \mid x$ and y are any strings of 0's and 1's$\}$

Conditions

1) Has it already seen 01? Yes, Then it will be in accepting state q_1
2) Has it never seen 01 & but its most recent input was 0
3) Has it never seen 01, but it last saw a 1 or, just started. q_0

Transition Diagram

- For each state there is a node
- For each state q in Q and each input symbol a in Σ, let $\delta(q, a) = p$. Then There is an arc from node q to node p labeled a.
- An arrow into the start state, labelled start.
- Nodes corresponding to accepting states marked by double circles.



Transition Table

	0	1
→ q_0	q_2	q_0
* q_1	q_1	q_1
q_2	q_2	q_1

- Rows of the table correspond to the states
- Columns corresponds to the input
- Entry for the rows corresponding to the state q and the column corresponding to input a is the state $\delta(q, a)$
- Start state marked with arrow
- Accepting states marked with Star.

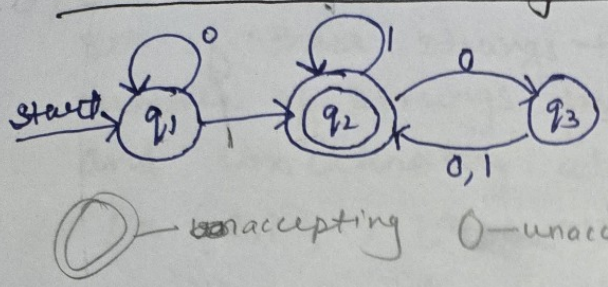
Q. $A = \{w | w$ contains at least one 1 and an even number of 0's follow the last 1$\}$

transition function

$M_1 = (\{q_1, q_2, q_3\}, \{0,1\}, \delta, q_1, \{q_2\})$ .1

$L(M_1) = A$ ex— 1, 1001, 10000

Transition or State diagram



Transition Table

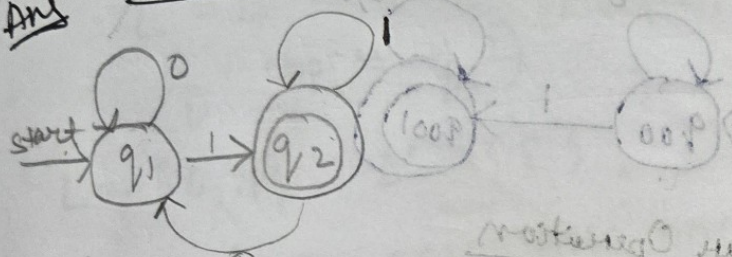
	0	1
→ q_1	q_1	q_2
* q_2	q_3	q_2
q_3	q_2	q_2

1—accepting 0—unaccepting

$M_2 = (\{q_1, q_2\}, \{0,1\}, \delta, q_1, \{q_2\})$

Q $L(M_2) = \{w \mid w \text{ ends in a } 1\}$

Ans Transition diagram



Transition Table

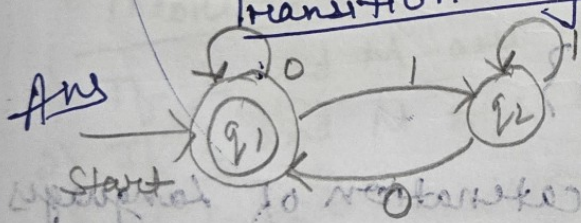
	0	1
→ q_1	q_1	q_2
* q_2	q_1	q_2

Q $L(M_3) = \{w \mid w$ is the empty string ε or ends in a $0\}$

$\Sigma = \{0,1\}$

Transition diagram for M_3

Ans

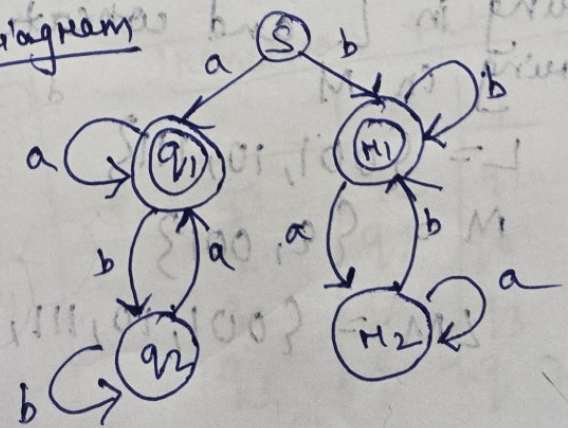


Transition Table

	0	1
* q_1	q_1	q_2
q_2	q_1	q_2

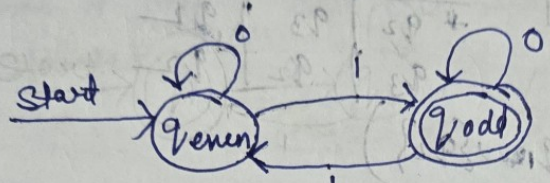
Accept all string that start and end with a or start and end with b

$\Sigma = \{a, b\}$ Transition Diagram



Q Accept all string with an odd number of 1's

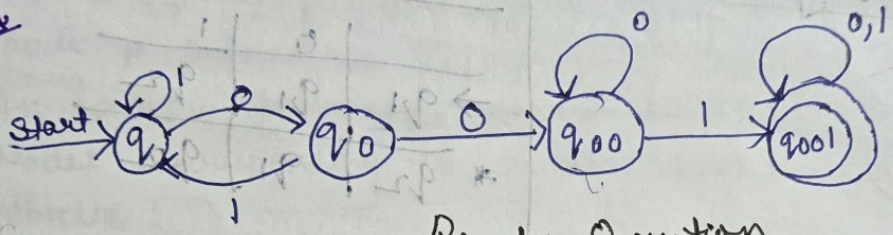
$\Sigma = \{0, 1\}$



Q Accept all strings that contains the string 001 as substring

$\Sigma = \{0, 1\}$

Ans



Regular Operation

1) $L \cup M$ ← union of two languages L and M set of strings that are either in L or M or both

$L = \{001, 10, 111\}$

$M = \{\varepsilon, 001\}$

$L \cup M = \{\varepsilon, 10, 001, 111\}$

2) $LM = L \cdot M$ ← concatenation of languages L and M

Set of strings that can be formed by taking any string in L and concatenating it with any string in M

$L = \{001, 10, 111\}$

$M = \{\varepsilon, 001\}$

$LM = \{001, 10, 111, 001001, 10001, 111001\}$

3) L^* ← Star or closure of Kleene closure of a language L
set of those strings that can be formed by taking any number of strings from L, possibly with repetitions and concatenating all of them → any combination

$L = \{0,1\}$, L^* ← all strings of 0's and 1's

$L = \{0, 11\}$ L^* ← all strings of 0's and 1's such that 1's come in pairs, eg, 011, 11110, ε,

L^* is the infinite union $\bigcup_{i \geq 0} L^i$, where $L^0 = \{\varepsilon\}$ $L' = L$ and L^i for $i > 1$ is $LL \cdots L$ (concatenation of i copies of L)

$L = \{0, 11\}$ $L^0 = \{\varepsilon\}$, $L' = \{0, 11\}$, $L^2 = \{00, 011, 110, 1111\}$

$L^3 = \{000, 0011, 0110, 1100, 01111, 11011, 11110, 111111\}$

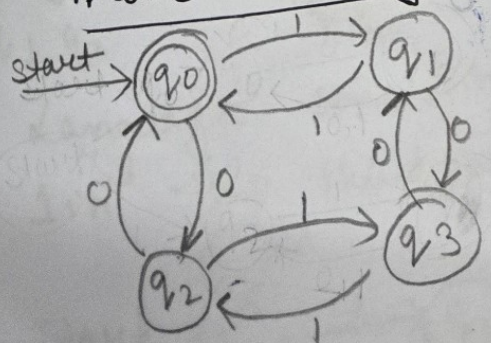
To compute L^*, we must compute L^i for each i and take the union of all these languages.

Theorems →

1) If L and M are regular languages, Then so is $L \cup M$
2) If L and M are regular languages, Then so is LM
3) If L is a regular languages, Then so is L^*.

$L = \{w \mid w$ has both an even number of 0's and an even number of 1's$\}$ $A = (\{q_0, q_1, q_2, q_3\}, \{0,1\}, \delta, q_0, \{q_0\})$

Transition Diagram



Transition table

	0	1
*→ q_0	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

Non-deterministic Finite Automata (NFA)

- Power to be in several states at once
- Accept the regular languages

Defination of NFA

$A = (Q, \Sigma, \delta, q_0, F)$

$Q \leftarrow$ finite set of states

$\Sigma \leftarrow$ finite set of input symbols

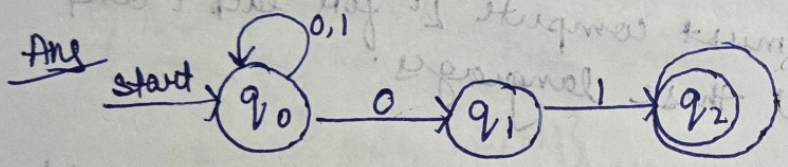
$q_0 \leftarrow$ start state, $q_0 \in Q$

$F \leftarrow$ set of final or accepting states, $F \subseteq Q$

$\delta \leftarrow$ transition function

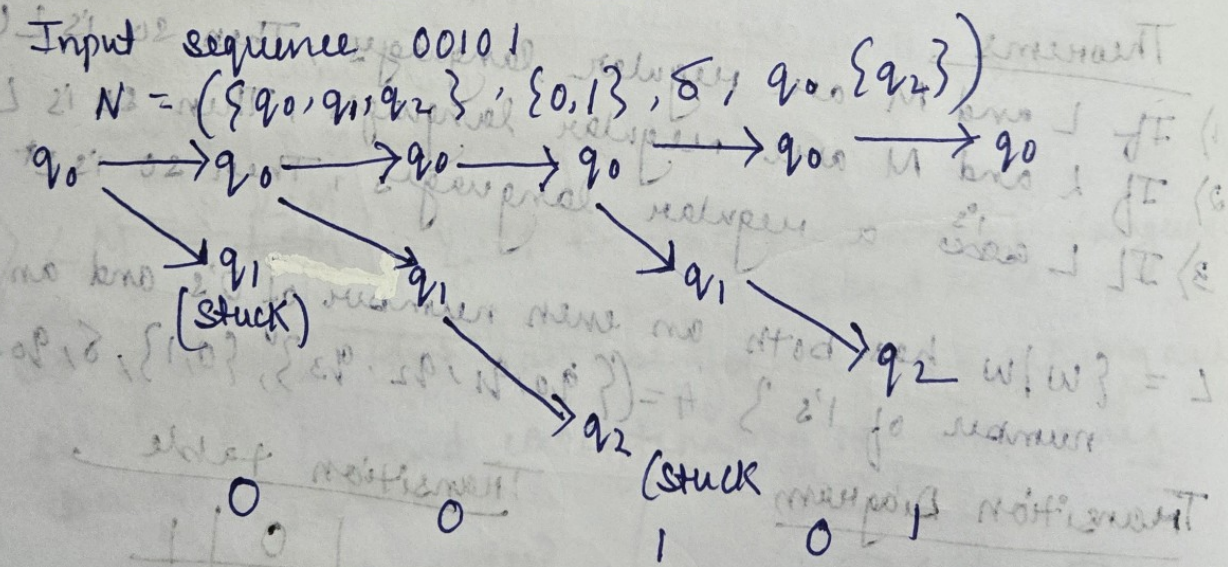
$$\delta : Q \times \Sigma \rightarrow 2^Q$$

Accept all strings that end in 01



Input sequence 00101

$N = (\{q_0, q_1, q_2\}, \{0,1\}, \delta, q_0, \{q_2\})$



	0	1
→ q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	ϕ	$\{q_2\}$
* q_0	ϕ	ϕ

01

ε-NFA Non

$A = (Q, \Sigma, \delta, q_0, F)$

All components have ~~the same~~ interpretation as for an NFA, except for δ

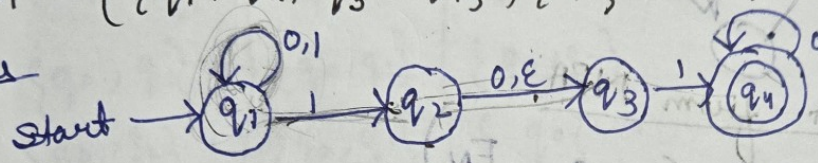
$$\delta : Q \times (\Sigma \cup \{\varepsilon\}) \longrightarrow 2^Q$$

Note, NFA and ε-NFA are different but in our ~~no~~ book the author has made them same so in midsem we have to see the question and understand

Q N, accept all strings that contain either 101 or 11 as substring

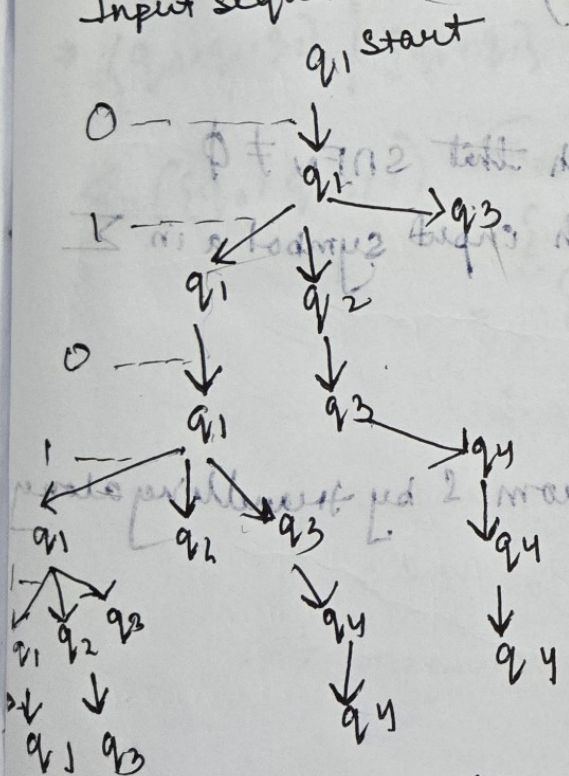
$N_1 = (\{q_1, q_2, q_3, q_4\}, \{0,1\}, \delta, q_1, \{q_4\})$

Ans

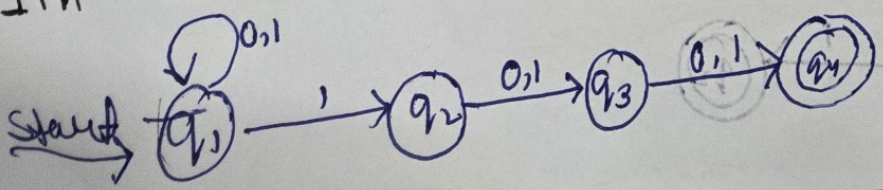


	0	1	ε
→ q_1	$\{q_1\}$	$\{q_1, q_2\}$	ϕ
q_2	$\{q_3\}$	ϕ	$\{q_3\}$
q_3	ϕ	$\{q_4\}$	ϕ
q_4	$\{q_4\}$	$\{q_4\}$	ϕ

Input sequence 01 0110



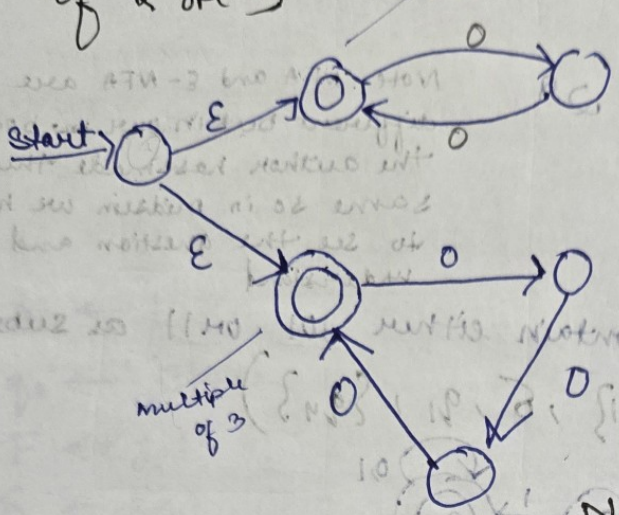
Language consisting of all strings over $\{0,1\}$ containing 1 in the third position from the end



Minm string

{100, 101, 110, 111, }

Accept all strings of the form 0^k where k is a multiple of 2 or 3



Construction of DFA from NFA

NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$

DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_N)$

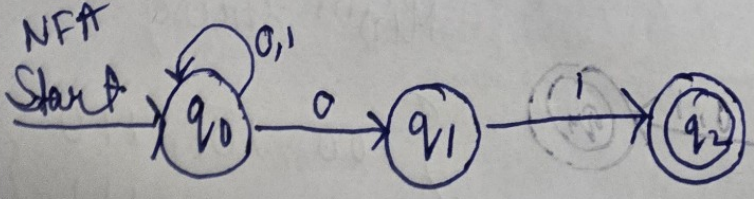
- $Q_D \leftarrow$ set of subsets of Q_N
- $F_D \leftarrow$ set of subsets S of Q_N such that $S \cap F_N \neq \phi$
- For each set $S \subseteq Q_N$ and for each input symbol a in Σ

$$\delta_D(S,a) = \bigcup_{p \text{ in } S} \delta_N(p,a)$$

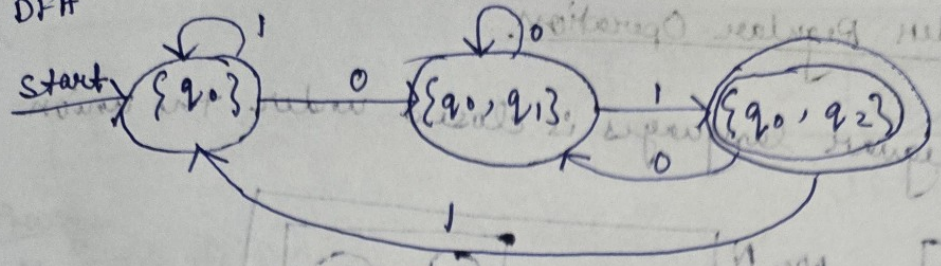
For an ε-NFA

$E(S) = \{q \mid q$ can be reached from S by traveling along 0 or more ε arrows$\}$

- start state $\leftarrow E(\{q_0\})$
- $\delta_D(S,a) = \bigcup_{p \text{ in } S} E(\delta_N(p,a))$



DFA



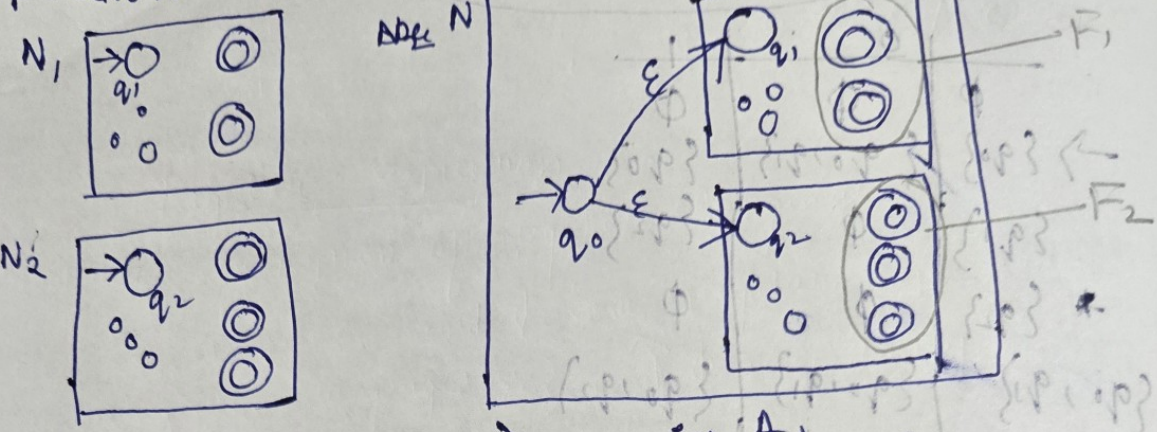
	0	1
ϕ	ϕ	ϕ
→ $\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_1\}$	ϕ	$\{q_2\}$
* $\{q_2\}$	ϕ	ϕ
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
* $\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$
* $\{q_1, q_2\}$	ϕ	$\{q_2\}$
* $\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$

$$\delta_D(\{q_0, q_1\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_1, 1)$$
$$= \{q_0\} \cup \{q_2\}$$
$$= \{q_0, q_2\}$$

Closure under Regular Operation

Theorem

The class of regular languages is closed under the union operation



Proof: $N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1

$N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2

Construct $N = (Q_1, \Sigma, \delta, q_0, F)$ to recognize $A_1 \cup A_2$

$Q = \{q_0\} \cup Q_1 \cup Q_2$

$q_0 \leftarrow$ start state of N

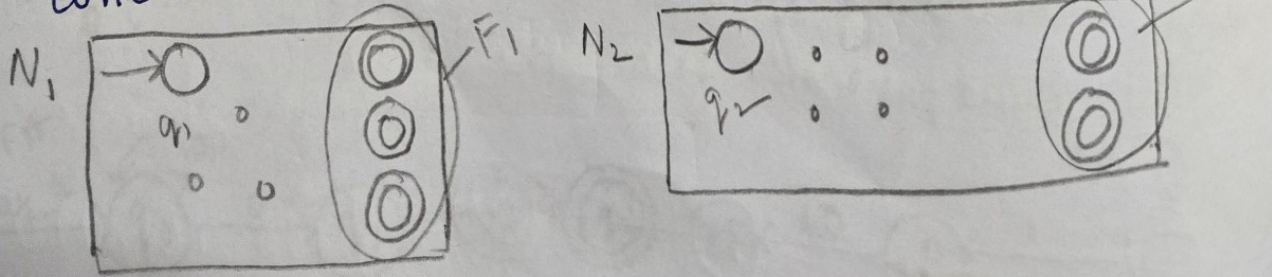
$F = F_1 \cup F_2$

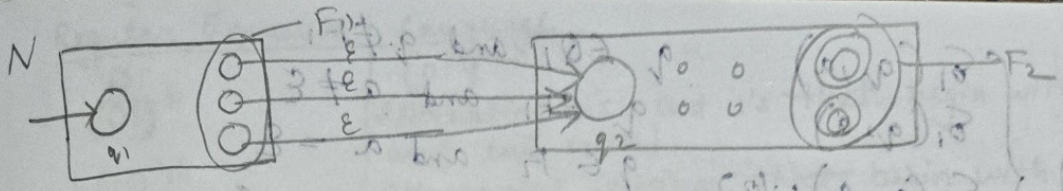
Define δ for any $q \in Q$ and any $a \in (\Sigma \cup \{\varepsilon\})$

$$\delta(q,a) = \begin{cases} \delta_1(q_0,a) & q \in Q_1 \\ \delta_2(q,a) & q \in Q_2 \\ \{q_1, q_2\} & q = q_0 \text{ and } a = \varepsilon \\ & q = q_0 \text{ and } a \neq \varepsilon \end{cases}$$

Theorem

The class of regular languages is closed under the concatenation operation





Proof

$N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1

$N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2

construct $N = (Q, \Sigma, \delta, q_1, F_2)$ to recognize A_1, A_2

$Q = Q_1 \cup Q_2$

$q_1 \leftarrow$ ~~start state~~ of N same as start state of N_1

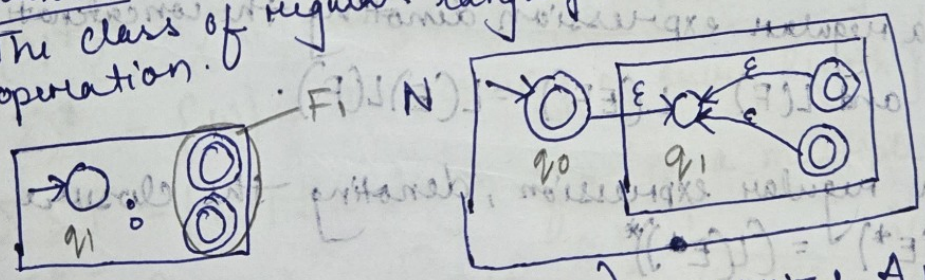
$F_2 \leftarrow$ accept states of N same as same as accept states of N_2

Define δ so that for any $q \in Q$ and any $a \in (\Sigma \cup \{\varepsilon\})$

$$\delta(q,a) = \begin{cases} \delta_1(q,a) & q \in Q_1 \text{ and } q \notin F_1 \\ \delta_1(q,a) & q \in F_1 \text{ and } a \neq \varepsilon \\ \delta_1(q,a) \cup \{q_2\} & q \in F_1 \text{ and } a = \varepsilon \\ \delta_2(q,a) & q \in Q_2 \end{cases}$$

Theorem

The class of regular languages is closed under the star operation.



Proof

$N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1

Construct $N = (Q, \Sigma, \delta, q_0, F)$ to recognize A_1^*

$Q = \{q_0\} \cup Q_1$

$q_0 \leftarrow$ start state of N

$F = \{q_0\} \cup F_1$

Define δ so that for any $q \in Q$ and any $a \in (\Sigma \cup \{\varepsilon\})$

$$\delta(q,a) \begin{cases} \delta_1(q,a) & q \in Q_1 \text{ and } q \notin F_1 \\ \delta_1(q,a) & q \in F_1 \text{ and } a \neq \varepsilon \\ \delta_1(q,a) \cup \{q_1\} & q \in F_1 \text{ and } a = \varepsilon \\ \{q_1\} & q \in q_0 \text{ and } a = \varepsilon \\ \phi & q = q_0 \text{ and } a \neq \varepsilon \end{cases}$$

Regular Expression

Algebraic description of regular languages

- The constants ε and ϕ are regular expressions, denoting the languages $\{\varepsilon\}$ and ϕ respectively. $L(\varepsilon) = \{\varepsilon\}$, $L(\phi) = \phi$
- If a is any symbol, Then a is a regular expression, denoting the language $\{a\}$. $L(a) = \{a\}$.
- If E and F are regular expressions, Then E+F or EUF is a regular expression denoting The union of L(E) and L(F). $L(E+F) = L(E) \cup L(F)$
- EF is a regular expression denoting the concatenation of L(E) and L(F). $L(EF) = L(E)L(F)$.
- E^* is a regular expression, denoting the closure of L(E). $L(E^*) = (L(E))^*$
- (E), parenthesized E is a regular expression, denoting the same language as E. $L((E)) = L(E)$

Q) Write a regular expression for the set of strings that consists of alternating 0's and 1's

Regular Expression	Language
01	{01}
(01)*	Alternating 0's and 1's that begin with 0 and end with 1
(10)*	Alternating strings that begin with 1 and end with 0
0(10)*	Strings that both begin and end with 0
1(01)*	Strings that both begin and end with 1

$(01)^* + (10)^* + 0(10)^* + 1(01)^* = (\varepsilon+1)(01)^*(\varepsilon+0)$

Precedence

Star operation – Highest
Concatenation operation – Next
Union operation – Lowest

Parenthesis can be used to groups operands exactly as we choose

$\Sigma = \{0,1\}$

$0^*10^* = \{w \mid w$ contains a single $1\}$

$\Sigma^*1\Sigma^* = \{w \mid w$ has atleast one $1\}$

$\Sigma^*001\Sigma^* = \{w \mid w$ contains the string 001 as substring$\}$

$1^*(01^+)^* = \{w \mid$ every 0 in w is followed by atleast one 1$\}$

$(\Sigma\Sigma)^* = \{w \mid w$ is a string of every length$\}$

$(\Sigma\Sigma\Sigma)^* = \{w \mid$ the length of w is a multiple of 3$\}$

$01 \cup 10 = \{01, 10\}$

$0\Sigma^*0 \cup 1\Sigma^*1 \cup 0 \cup 1 = \{w \mid w$ starts and ends with the same symbol$\}$

$(0 \cup \varepsilon)1^* = 01^* \cup 1^*$

$(0 \cup \varepsilon)(1 \cup \varepsilon) = \{\varepsilon, 0, 1, 01\}$

$1^*\phi = \phi$,

$\phi^* = \{\varepsilon\}$

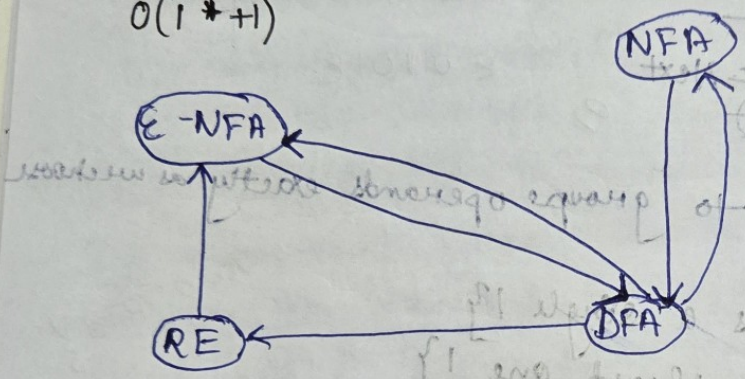
Q 01*+1

- 1 — union
- * — star
- 0 — concatination

String 1 plus all string consisting of a 0 followed by any number of 1's (including none).

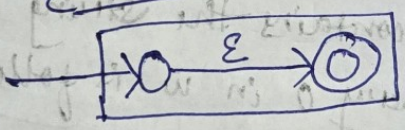
Q (01)*+1

0(1*+1)

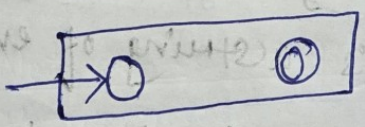


RE

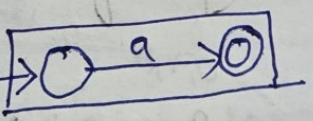
ε-NFA

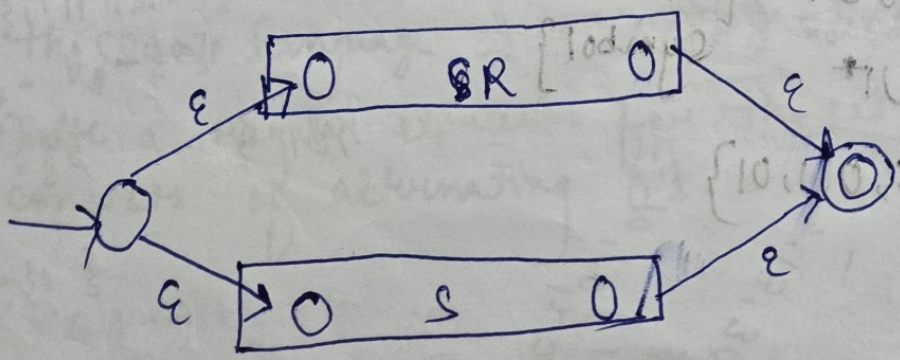


ϕ

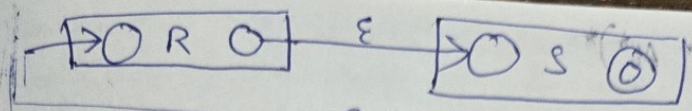


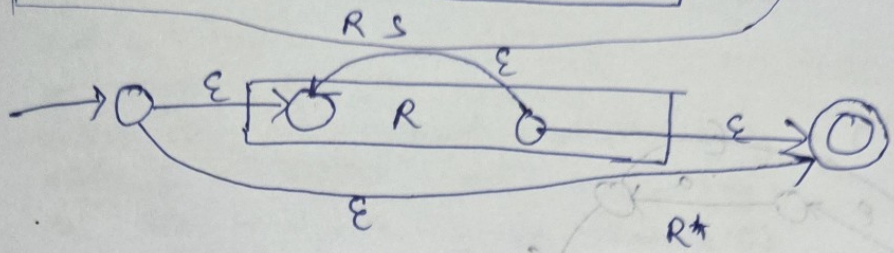
a



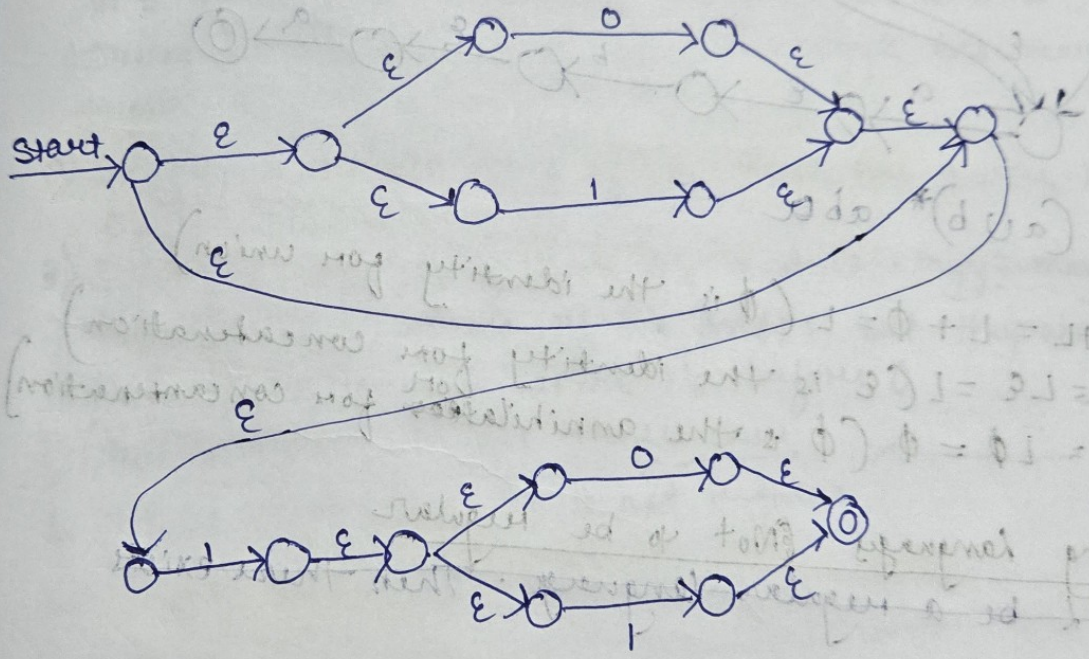


R+S

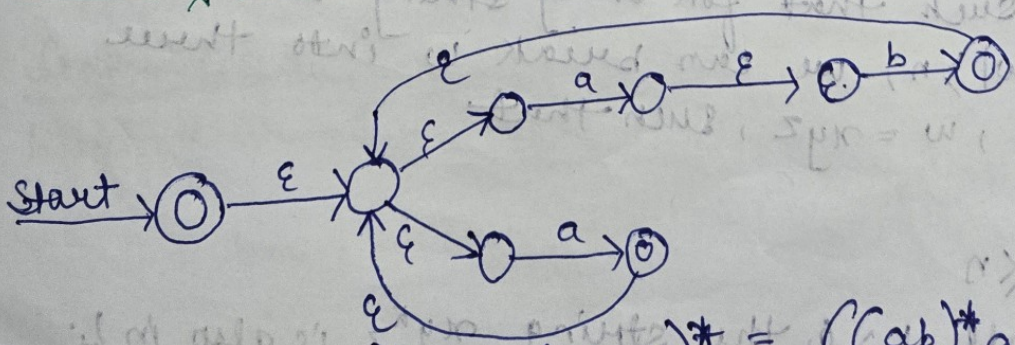




Automation of R also serves as the automaton of (R)

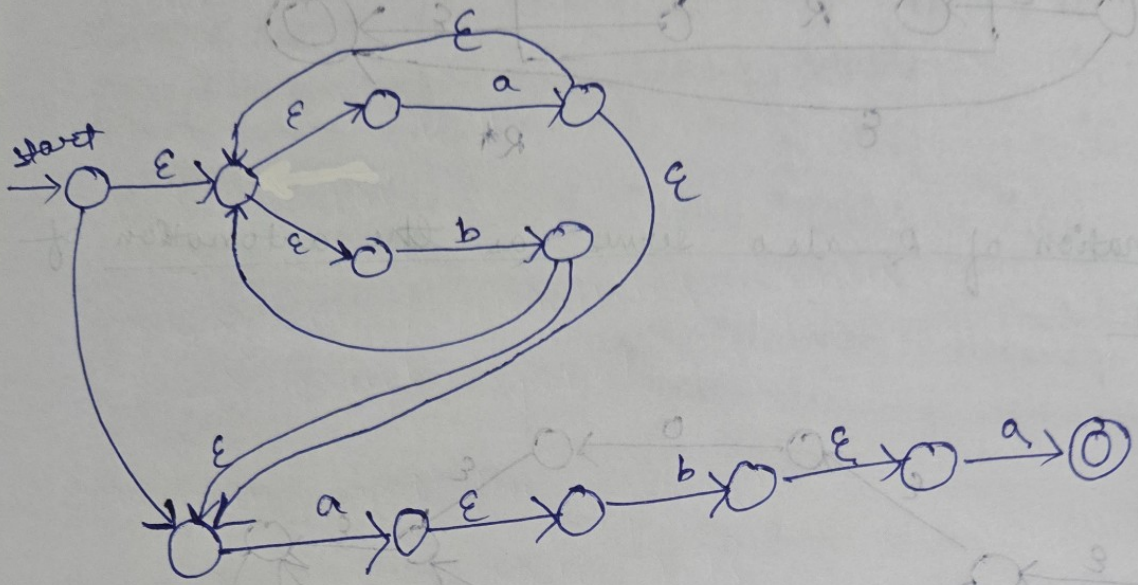


$(0+1)^* 1 (0+1)$



RE: $(ab \cup a)^* = (ab+a)^* = ((ab)^* a^*)^*$

$(L+M)^* = (L^* M^*)^*$



$(a \cup b)^* abba$

$\phi + L = L + \phi = L$ (ϕ is the identity for union)
$\varepsilon L = L\varepsilon = L$ (ε is the identity for concatenation)
$\phi L = L\phi = \phi$ (ϕ is the annihilator for concatenation)

Proving Languages Not to be regular

~~Let L be a regular language. Then there exists~~

Pumping Lemma

Let L be a regular language. Then there exists a constants n (called the pumping length which depends on L) such that for every string w in L such that $|w| \geq n$, we can break w into three strings, $w = xyz$, such that

1) $y \neq \varepsilon$
2) $|xy| \leq n$
3) For all $K > 0$, the string $xy^K z$ is also in L.

Languages L_{eq} consisting of all strings with an equal number of 0's and 1's is not regular

Let $w = 0^n 1^n$

$w = xyz$, $y \neq \varepsilon$

Since $|xy| \leq n$ x and y consist only of 0's

when $k=0$, $xy^k z = xz$ is in L_{eq}.

xz has n 1's and fewer than n 0's

Contradiction

$L = \{0^n 1^n / n \geq 0\}$ is not regular.

$w = 0^p 1^p$ where p is the pumping length

$w = xyz$, for any $k \geq 0$, $xy^k z$ is in L

1) String y consists only of 0's. xyyz has more 0's than 1's

2) String y consists only of 1's. xyyz has more 1's than 0's

3) String y consists of both 0's and 1's. xyyz may have the same number of 0's and 1's but they will be out of order with some 1's before 0.

Contradiction in all three cases

$L = \{ss / s \in \{0,1\}^*\}$ is not regular

$w = 0^n 1 0^n 1$, $s = 0^n 1$

$w = xyz$

$xyyz \notin L$, because y must consists only of 0's as $|xy| \leq n$

Contradiction

What if $w = 0^n 0^n$, $s = 0^n$?